

Lab 7: Data Visualization and Rmarkdown

In this lab, you will use data from the 2019 National Survey of Early Care and Education (NSECE) Workforce Questionnaire to practice data preparation, data visualization, and report generation using R and R Markdown.

1. Open a .R file and set up your working environment

Load the following packages: "tidyverse", "ggplot2", "knitr", "rmarkdown", "scales", "stargazer"

2. Use the .R file to prepare the data for analysis

Subset the NSECE dataset to retain the following variables:

- WF9_WORK_FT – Staff full-time or part-time employment status
- WF9_CHAR_GENDER – Staff gender
- WF9_CAREER_EXPERIENCE – Years of experience caring for children under age 13
- WF9_CHAR_RACE – Staff race
- WF9_WORK_WAGE – Hourly wage

Rename each variable using clear, descriptive names. For each variable, identify its measurement level (nominal, ordinal, or interval/ratio) and indicate whether it is discrete or continuous.

Next, create the following derived variables:

- Full-time: 1 if the staff member worked full time; 0 otherwise.
- Female: 1 if the staff member identified as female; 0 otherwise.
- Create k-1 binary variables based on years of experience working with children (k = number of categories). After examining the variable, choose a reference group.
- Create k-1 binary variables based on race (k = number of categories). After examining the variable, choose a reference group.
- Transform hourly wage to a logarithm.

3. Once your data preparation is complete, write the code needed to generate and save an HTML report.

4. Open a .Rmd file to create an HTML document

Include the following:

- Title: Lab 7
- Author: Your name

- Date
- Table of contents
- Introduction

5. Summarize individual variables

Using the original variables (not the dummy variables):

- Calculate the appropriate measures of central tendency, dispersion, and shape (when applicable).
- Interpret each statistic.
- Create the most appropriate visualization for each variable.

6. Explore associations

Using the original variables (not the dummy variables), create appropriate visualizations for the following relationships:

- Hourly wage by race.
- Years of career experience by gender.

7. Run a linear regression model

Estimate the following OLS regression model.

$$\begin{aligned} \log(\text{Hourly Wage}) &= \beta_0 + \beta_1(\text{Full Time}) + \beta_2(\text{Female}) + \beta_3 \dots \beta_7(\text{Years of Experience categories}) \\ &+ \beta_{16}(\text{Race categories}) + \varepsilon \end{aligned}$$

Include the following:

- Display the regression results in a well-formatted table.
- Create a residuals plot.
- Create a histogram of the model residuals.
- Based on these plots, discuss whether the assumptions of linearity, homoscedasticity, and normality appear to be satisfied.
- Interpret the coefficient estimates for each independent variable.
- Interpret the overall model fit using R^2 and adjusted R^2 .

8. Render the HTML report

- Re-run the entire .R script to ensure all code executes without errors.
- “Knit” the .Rmd file to generate the HTML report.
- Review the HTML output for formatting, accuracy, and completeness, and make any necessary revisions.

